DATASCI 406 FA 26, Quiz 2.2

Name	Uniqname	UMID

Each question is worth 1 point. Circle one answer for each.

Question 1 (1 point)

Which line draws `n` values from the 50/50 mixture of $N(-2, 1)$ and $N(2, 1)$?

1. `0.5 * rnorm(n, -2) + 0.5 * rnorm(n, 2)`
2. `rnorm(n, mean = sample(c(-2, 2), n, replace = TRUE))`
3. `rnorm(n, mean = 0.5 * (-2) + 0.5 * 2)`
4. `sample(c(rnorm(1, -2), rnorm(1, 2)), n, replace = TRUE)`

Solution

Answer: 2. This is the random-parameter reading: each draw first picks its mean, then draws from that normal. Option 1 averages the two variables, which gives the unimodal $N(0, 1/2)$; a sum is not a mixture. Option 3 averages the means and draws from $N(0, 1)$. Option 4 draws each component only once, so all `n` values are one of two numbers.

Question 2 (1 point)

X has density $f(x) = \frac{3}{2}x^2$ on $-1 \le x \le 1$. On $[0, 1]$, $X \mid X > 0$ has CDF x^3. Which line draws `n` values of X?

1. `runif(n, -1, 1)^(1/3)`
2. `sample(c(-1, 1), n, replace = TRUE) * runif(n)^(1/3)`
3. `sample(c(-1, 1), n, replace = TRUE) * runif(n)^3`
4. `2 * runif(n)^(1/3) - 1`

Solution

Answer: 2. f is symmetric, so $P(X > 0) = 1/2$, and $X \mid X > 0$ has quantile function $u^{1/3}$. A random sign times a draw of $X \mid X > 0$ is the Laplace construction from the video. Option 3 applies the CDF instead of its inverse. Option 4 rescales a $[0, 1]$ variable to $[-1, 1]$, which does not preserve the density. Option 1 takes cube roots of negative numbers, which R returns as `NaN`, and on $[0, 1]$ it would not give density $3x^2$ anyway.

Question 3 (1 point)

X and Y are independent $U(0, 1)$ and $Z = X + Y$. Which is true?

1. $Z \sim U(0, 2)$, since adding two uniforms gives a uniform.
2. Z is a 50/50 mixture of $U(0, 1)$ and $U(1, 2)$.
3. Z has density z on $[0, 1]$ and $2 - z$ on $[1, 2]$.
4. $Z \sim N(1, 1/6)$ exactly, by the central limit theorem.

Solution

Answer: 3. The convolution $\int f_X(z - y) f_Y(y)\, dy$ is the length of the overlap of $[0, 1]$ and $[z - 1, z]$, a triangle peaking at $z = 1$. Sums near 1 can be made many ways and sums near 0 or 2 only one way, so option 1 is wrong. Option 2 has density $1/2$ everywhere on $[0, 2]$, which is option 1 again. Option 4 has the right mean and variance but is a large-sample approximation, and Z is bounded.

Question 4 (1 point)

Under $H_0 : \mu = 0$, data are drawn from the mixture $0.8\, N(0, 1) + 0.2\, N(0, 10)$ (variances), and the test rejects when $|\bar{X}| \ge 1.96/\sqrt{n}$. Why does it reject more than 5% of the time?

1. The contamination moves the mean of the data away from 0.

2. Each observation has variance $0.8 \cdot 1 + 0.2 \cdot 10 = 2.8$, so $\bar{X}$ is more spread out than the cutoff assumes.
3. The sample mean is a biased estimator under contamination.
4. n is too small; with $n = 1000$ the size would be 0.05.

Solution

Answer: 2. Both components have mean 0, so the mixture has mean 0 and variance $\sum a_i \sigma_i^2 = 2.8$. The cutoff assumes variance 1. Options 1 and 3 are wrong for the same reason: $E(\bar{X}) = 0$. Option 4 is wrong because the cutoff uses the wrong variance at every n; the size does not improve as n grows.

Rubric. 1 pt per question, no partial credit.